MSBA 7023: Geospatial and Business Analytics (December 2025- January 2026)

COURSE SYLLABUS

COURSE DESCRIPTION

Location, concerning the "where" aspect of all businesses, is important everywhere: from traditional industries such as, agriculture, banking, real estate, utility services, to modern e-retailing, healthcare, and location-based service platforms like ride-sharing and urban deliveries. Thanks to the overwhelming growth of online purchasing, the penetration of smartphones and locate-aware apps, and the looming prospect of Internet-Of-Things, "every single new 21st century data source contains location (ESRI)." In such background, companies increasingly look for business analysts with geospatial mindset and toolbox to unlock opportunities of growth and better customer services. This course introduces the students to the basic knowledge of geospatial data, system, methods, and helps them develop mindset and toolbox to tackle challenging business problems related to location.

This course aims to equip students with basic knowledge of Geographic Information Systems (GIS) and business analytical skills to analyze data with geospatial tags and features, including skills of integration and manipulation of data layers, descriptive analysis (e.g., visualize business data on well-designed maps to reveal patterns and inform decisions), predictive analysis (e.g., make good use of spatial features to better forecast growth and cost), and prescriptive analysis (e.g., facility location evaluation, routing & logistics planning, targeted marketing). It provides the students with problem-solving experience with "classic" and modern applications in various business contexts, so that they can develop mindset and toolbox of geospatial analysis.

INSTRUCTOR

Professor Zhixi Wan

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LEARNING MATERIALS

- Handouts and assignments will be posted on Moodle

ASSESSMENT

Class Participation (20%) (Individual)

- Class attendance, discussion, presentation and playing a proactive role in other in-class activities.

Assignment (50%)

- Work on assigned problems and answer certain questions (40%) (Individual)
- In-class mini case competition (10%) (Group)

Group project (30%) (Group)

- Project topics will be proposed by project members (with lecturer's approval).
- You need to report your project by a StoryMap.

Students need to form groups for the project.

COURSE POLICIES - Academic Dishonesty

The University Regulations on academic dishonesty will be strictly enforced. Please check the University Statement on plagiarism at http://www.hku.hk/plagiarism/

Where a candidate for a degree or other award uses the work of another person or persons without due acknowledgement:

- The relevant Board of Examiners may impose a penalty in relation to the seriousness of the offence;
- The relevant Board of Examiners may report the candidate to the Senate, where there is prima facie evidence of an intention to deceive and where sanctions beyond those in (a) might be invoked.

LECTURE SCHEDULE (Tentative)

Lecture		Subject	Demos, Assignments, and Optional	Due
1	09DEC 630PM	***Introduction*** ▪ Geographic Information System ▪ Spatial Data and Various formats of spatial file ▪ Business application examples	▪ Demo: Create a map of NYC bank locations ▪ Optional: Ten big ideas about applying the Science of Where ▪ Todo: Form Your Project Group	
2	12DEC 630PM	***Tools*** ▪ Create and manage data layers ▪ ArcGIS Python API	▪ Demo: Publish a data layer ▪ Demo: Spatial analysis of NYC taxi data ▪ Assignment 1: Explore NYC taxi data	
3	16DEC 630PM	***Visualization*** ▪ Use the SmartMap capability in ArcGIS Online ▪ Visualize patterns, relationship, and changes	▪ Demo: SmartMap examples ▪ Optional reading: Working with temporal data	Project Group Formation
4	19DEC 630PM	***Spatial Analysis: Intro*** ▪ Spatial measurements, statistics, and distributions ▪ Overview of spatial analysis ▪ Hotspot, outlier, clustering, geo-enrichment	▪ Assignment 2: Explore commercial permits	Assignment 1 Project Proposal
5	02JAN 630PM	***Spatial Analysis: Patterns*** ▪ Buffering ▪ Service areas ▪ Summarize proximity	▪ Demo: Solving a spatial problem	
6	06JAN 630PM	***Making Predictions*** ▪ Point interpolation ▪ Regression model for spatial data	▪ Demo: Point interpolation for air quality ▪ Assignment 3: Spatial analysis and predictions of house prices	Assignment 2
7	09JAN 630PM	***Site Evaluation and Choice*** ▪ Overlay analysis ▪ Site suitability models	▪ Demo: Site suitability for a windmill farm ▪ HKTVmall case study	Project Progress Report
8	13JAN 630PM	***Network and Routing*** ▪ Transportation network ▪ Proximity analysis ▪ Routing analysis ▪ Location allocation	▪ Demo: Delivery logistics using the VRP solver ▪ Demo: Choosing best facilities for health clinics	Assignment 3
9	16JAN 630PM	***Applications & Case Prep*** ▪ Bank location optimization ▪ Geospatial data in ride-sharing business ▪ Analysis of demand and supply	▪ Case Assignment: Amazon "last-mile" distribution network in New York City	
10	20JAN 630PM	***Course Wrap-up*** ▪ Case competition & discussion ▪ Wrap-up		Assignment 4

REFERENCES (OPTIONAL)

- The ArcGIS Book: 10 Big Ideas about Applying the Science of Where. Available on Moodle; also here: https://learn.arcgis.com/en/arcgis-book/
- Location Analytics: The Key to More Powerful Analysis. ESRI. Available on Moodle
- Free online learning resources at https://learn.arcgis.com/en/gallery/